

But the work is precisely the *electric potential energy* U of the system $q_1 + q_2$. Thus

$$U (= W) = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_{12}}$$

The subscript of r emphasizes that the distance involved is that between the point charges q_1 and q_2 .

For a system containing more than two charges, the procedure is to compute the potential energy for every pair of charge separately. The results are then added separately. This procedure rests on a physical picture in which (i) charge q_1 is brought into position, (ii) q_2 is brought from infinity to its position near q_1 , (iii) q_3 is brought from infinity to its position near q_1 and q_2 , etc.

Example 3.4 What must the magnitude of an isolated positive point charge be for the electric potential at 10 cm from the charge to be +100 volts?

Soln.

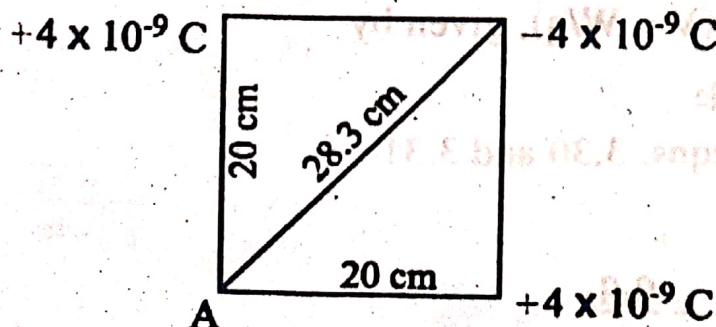
From $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$ we obtain

$$q = (V) (4\pi\epsilon_0) (r)$$

$$= (100 \text{ volts}) (4\pi) (8.9 \times 10^{-12} \text{ coul}^2/\text{nt-m}^2) (0.10 \text{ m})$$

$$= 1.1 \times 10^{-9} \text{ coul.}$$

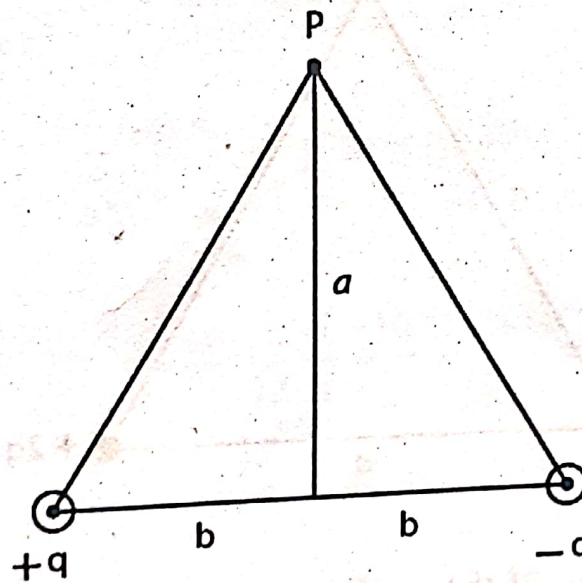
Example 3.5 Three charges are placed at three corners of a square as shown in the figure. Find the potential at point A.



Soln.

$$\begin{aligned}
 V_A &= \frac{1}{4\pi\epsilon_0} \left[\frac{+4 \times 10^{-9} \text{ C}}{0.20 \text{ m}} + \frac{-4 \times 10^{-9} \text{ C}}{0.283 \text{ m}} + \frac{+4 \times 10^{-9} \text{ C}}{0.20 \text{ m}} \right] \\
 &= (9 \times 10^9) \left[\frac{4 \times 10^{-9} \text{ C}}{0.20 \text{ m}} + \frac{-4 \times 10^{-9} \text{ C}}{0.283 \text{ m}} + \frac{4 \times 10^{-9} \text{ C}}{0.20 \text{ m}} \right] \\
 &= 233 \text{ V.}
 \end{aligned}$$

Example 3.6 Show that the absolute potential at point P in the figure below is zero.



Soln.

$$\begin{aligned}
 V_P &= \frac{1}{4\pi\epsilon_0} \frac{+q}{\sqrt{a^2 + b^2}} + \frac{1}{4\pi\epsilon_0} \frac{-q}{\sqrt{a^2 + b^2}} \\
 &= 0.
 \end{aligned}$$

Example 3.7 Two protons in a nucleus of U^{238} are 6.0×10^{-15} metre apart. What is their mutual electric potential energy?

Soln.

$$U = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r} = \frac{(9.0 \times 10^9 \text{ nt} \cdot \text{m}^2 / \text{coul}^2) (1.6 \times 10^{-19} \text{ coul})^2}{6.0 \times 10^{-15} \text{ m}}$$

Example 3.9 Two metal spheres are 3.0 cm in radius and carry charges of $+1.0 \times 10^{-8}$ coul and -3.0×10^{-8} coul respectively, assumed to be uniformly distributed. If their centre are 2.0m apart, calculate (a) the potential of the point halfway between their centres and (b) the potential of each sphere.

Soln.

The charges may be assumed to be located at the centres of each sphere. The midway point will be 1.0 metre from either centres. The potential at the mid-point is then given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_1}{r_1} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r_2}$$

But $r_1 = r_2 = 1.0$ metre.

$$\begin{aligned} \therefore V &= \frac{1}{4\pi\epsilon_0} \left[\frac{q_1 + q_2}{r} \right] \\ &= \frac{(9.0 \times 10^9 \text{ N-m}^2/\text{c}^2)(1 \times 10^{-8} - 3 \times 10^{-8})}{1.0\text{m}} \\ &= \frac{-18 \times 10^9 \times 10^{-8}}{1.0} = -180 \text{ volts.} \end{aligned}$$

- (b) The potential at the surface of sphere 1 is due to its own charge q_1 plus that due to charge q_2 of sphere 2 at a distance r .

$$\begin{aligned} V &= \frac{1}{4\pi\epsilon_0} \frac{q_1}{R} + \frac{1}{4\pi\epsilon_0} \frac{q_2}{r} & R = 3 \text{ cm} = 0.03 \text{ m} \\ & & r = 2 \text{ m} \\ &= (9 \times 10^9 \text{ N-m}^2/\text{coul}^2) \left[\frac{1 \times 10^{-8} \text{ c}}{0.03} + \frac{-3 \times 10^{-8} \text{ c}}{2.0\text{m}} \right] \\ &= 2864 \text{ volts.} \end{aligned}$$

The potential at the surface of sphere 2 is due to its own charge q_2 plus that due to charge q_1 of sphere 1 at a distance r .

$$V = \frac{1}{4\pi\epsilon_0} \frac{q_2}{r} + \frac{1}{4\pi\epsilon_0} \frac{q_1}{R}$$

$$= (9 \times 10^9 \text{ N-m}^2/\text{coul}^2) \left[\frac{-3 \times 10^{-8} \text{ c}}{0.03 \text{ m}} + \frac{1 \times 10^{-8} \text{ c}}{2 \text{ m}} \right]$$

$$= -8955 \text{ volts.}$$

Example 3.10 Calculate (i) the electric potential established by the nucleus of a hydrogen atom at the mean distance of the circulating electron ($r = 5.3 \times 10^{-11} \text{ m}$), (ii) the electric potential energy of the atom when the electron is at its radius and (iii) the kinetic energy of the electron assuming it to be moving in a circular orbit of this radius centred on the nucleus. (iv) how much energy is required to ionize the hydrogen atom? Express all energies in electron-volts.

Soln.

(i) The electric potential is given by

$$V = \frac{1}{4\pi\epsilon_0} \frac{q}{r} = \frac{(9 \times 10^9 \text{ N-m}^2/\text{c}^2)(1.6 \times 10^{-19} \text{ c})}{5.3 \times 10^{-11} \text{ m}}$$

$$= 27.1 \text{ volts.}$$

(ii) Electric potential energy of the atom is given by

$$U = qV = -e.V = -(1.6 \times 10^{-19} \text{ c})(27.1 \text{ volts})$$

$$= - \frac{(1.6 \times 10^{-19} \text{ c})(27.1 \text{ volts})}{1.6 \times 10^{-19}} \text{ eV}$$

$$= -27.1 \text{ eV.}$$

(iii) From the relation of electrostatic force balancing centripetal force, we obtain

$$\frac{mv^2}{r} = \frac{1}{4\pi\epsilon_0} \frac{e^2}{r^2}$$

$$\text{or, } mv^2 = \frac{e^2}{4\pi\epsilon_0 r} = 27.1 \text{ eV}$$

$$\therefore \text{K. E.} = \frac{1}{2}mv^2 = 13.55 \text{ eV.}$$

$$\begin{aligned}
 \text{(iv) Total energy} &= \text{kinetic energy} + \text{potential energy} \\
 &= 13.55 \text{ eV} - 27.1 \text{ eV} \\
 &= -13.55 \text{ eV.}
 \end{aligned}$$

This is the energy required to ionize the hydrogen atom.

Example 3.11 A spherical drop of water carrying a charge of 3×10^{-6} coul has a potential of 500 volts at its surface (i) What is the radius of the drop? (ii) If two such drops of the same charge and radius combine to form a single spherical drop, what is the potential at the surface of the new drop formed?

Soln.

(i) From $V = \frac{1}{4\pi\epsilon_0} \frac{q}{r}$, we have

$$\begin{aligned}
 r &= \frac{1}{4\pi\epsilon_0} \frac{q}{V} = (9 \times 10^9 \text{ N-m}^2/\text{C}^2) \times \frac{3 \times 10^{-6} \text{ C}}{500 \text{ V}} \\
 &= 54 \text{ m.}
 \end{aligned}$$

(ii) Assuming the drops to be incompressible, the volume of the new drop will be twice the volume of the small drop.

$$\text{or, } \frac{4}{3}\pi R^3 = 2 \times \frac{4}{3}\pi r^3$$

where R is the radius of the new drop.

$$\therefore R = 2^{1/3} \cdot r$$

the charge on the new drop, $Q = 2q$.

$\therefore$ potential at the surface of the new drop

$$\begin{aligned}
 &= \frac{1}{4\pi\epsilon_0} \frac{Q}{R} = \frac{1}{4\pi\epsilon_0} \frac{2q}{2^{1/3} \cdot r} = \frac{1}{4\pi\epsilon_0} \frac{2^{2/3}q}{r} \\
 &= 500 \times 2^{2/3} \text{ volts} \\
 &= 790 \text{ volts.}
 \end{aligned}$$

Between the sheets, the two fields point to the same direction. They, therefore, add to give

$$E_c = 3.84 \times 10^5 \text{ N/C} + 2.43 \times 10^5 \text{ N/C} = 6.3 \times 10^5 \text{ N/C.}$$

Example 2.6. Two parallel plates of area 20 cm^2 each are separated by a distance of 10 mm . If one plate carries a charge $+2 \times 10^{-9} \text{ C}$, and the other carries a charge $-2 \times 10^{-9} \text{ C}$, calculate the electric field in the space between them.

Soln.

Surface charge density for each of the plate is given by

$$\sigma = \frac{2 \times 10^{-9}}{20 \times 10^{-4}} \text{ Cm}^{-2} = 10^{-6} \text{ Cm}^{-2}.$$

Since one of the plate carries positive charge while the other carries negative charge, the electric field in the space between them

$$E = \frac{\sigma}{\epsilon_0} = \frac{10^{-6} \text{ Cm}^{-2}}{8.85 \times 10^{-12} \text{ C}^2 \text{ N}^{-1} \text{ m}^{-2}} = 1.13 \times 10^5 \text{ N/C.}$$

Example 2.7 The electric strength of air is about $3.0 \times 10^6 \text{ N/C}$ which means that if the electric field exceeds this value, sparking will occur. What is the largest charge a 0.50 cm . radius sphere can hold if sparking is not to occur in the air surrounding it?

Soln.

For a point outside a sphere, all the charge on the surface can be considered to be located at the centre of the sphere. Thus the value of q can be obtained from the relation.

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2}$$

$$\text{or, } 3.0 \times 10^6 \text{ N/C} = \frac{1}{(4\pi)(8.85 \times 10^{-12} \text{ C}^2 / \text{N.m}^2)} \frac{q}{(5.0 \times 10^{-3} \text{ m})^2}$$

$$\text{or, } q = 8.34 \times 10^{-9} \text{ C} = 8.34 \text{ nC.}$$

$$U = -pE \cos\theta$$

This, again, can be put in the vector form

$$U = -\mathbf{p} \cdot \mathbf{E}$$

• Example 1.19 An electric dipole consists of two opposite charges of magnitude $2.0 \times 10^{-6} \text{C}$ separated by a distance $l = 1.0 \text{cm}$. It is placed in an external electric field of $2.0 \times 10^5 \text{N/C}$.

- (i) What maximum torque does the field exert on the dipole?
- (ii) How much work must an external agent do to turn the dipole from its initial alignment given by $\theta = 0^\circ$ to final alignment $\theta = 90^\circ$?

Soln.

- (i) The maximum torque is found by putting $\theta = 90^\circ$ in eqn. 1.19

$$\begin{aligned} \tau &= pE \sin\theta & q &= 2.0 \times 10^{-6} \text{C} \\ &= qlE \sin 90^\circ & l &= 1.0 \text{ cm} = 1 \times 10^{-2} \text{m} \\ & & E &= 2.0 \times 10^5 \text{N/C} \\ &= (2.0 \times 10^{-6} \text{C}) (1 \times 10^{-2} \text{m}) (2 \times 10^5 \text{N/C}) \\ &= 4 \times 10^{-3} \text{N.m.} \end{aligned}$$

- (ii) The work done is the difference in the potential energy between the positions $\theta = 90^\circ$ and $\theta = 0^\circ$.

$$\begin{aligned} W &= U_{90^\circ} - U_{0^\circ} = -pE \cos 90^\circ - (-pE \cos 0^\circ) \\ &= pE = qlE = 4 \times 10^{-3} \text{joules.} \end{aligned}$$

• Example 1.20 A molecule of water vapour (H_2O) has an electric dipole moment of magnitude $p = 6.2 \times 10^{-30} \text{C.m}$. The dipole moment arises because the effective centre of positive charge does not coincide with the effective centre of negative charge.

- (i) How far apart are the effective centres of positive and negative charge in a molecule of H_2O ?

(ii) What is the maximum torque on a molecule of H_2O in an electric field of magnitude $1.5 \times 10^4 \text{ N/C}$?

(iii) Suppose the dipole moment of a molecule of H_2O is initially pointing in a direction opposite to the field. How much work is done by the electric field in rotating the molecule into alignment with the field?

Soln.

(i) There are 10 electrons and, correspondingly, 10 positive charges in this molecule. We can write, for the magnitude of the dipole moment,

$$p = qd = (10e) d$$

where d is the separation between the centres of the positive and negative charges. e is the elementary charge. Thus

$$\begin{aligned} d &= \frac{p}{10e} = \frac{6.2 \times 10^{-30} \text{ C.m}}{(10)(1.6 \times 10^{-19} \text{ C})} \\ &= 3.9 \times 10^{-12} \text{ m} = 3.9 \text{ pm} \end{aligned}$$

(ii) As in eqn. 1.19, the torque is maximum when $\theta = 90^\circ$. Substituting this value in that equation yields,

$$\begin{aligned} \tau &= pE \sin \theta \\ &= (6.2 \times 10^{-30} \text{ C.m})(1.5 \times 10^4 \text{ N/C})(\sin 90^\circ) \\ &= 9.3 \times 10^{-26} \text{ N.m.} \end{aligned}$$

(iii) The work done in rotating the dipole from $\theta_0 = 180^\circ$ to $\theta = 0^\circ$ is given by eqn.

$$\begin{aligned} W &= \int_{\theta_0}^{\theta} pE \sin \theta \cdot d\theta \\ &= \left[-pE \cos \theta \right]_{180^\circ}^{0^\circ} \\ &= -pE \cos 180^\circ - (-pE \cos 0^\circ) \\ &= pE + pE \\ &= 2pE = (2)(6.2 \times 10^{-30} \text{ C.m})(1.5 \times 10^4 \text{ N/C}) \\ &= 1.9 \times 10^{-25} \text{ J.} \end{aligned}$$

This is the electric field produced by an infinite sheet of uniform charge, located on one side of a nonconductor such as plastic.

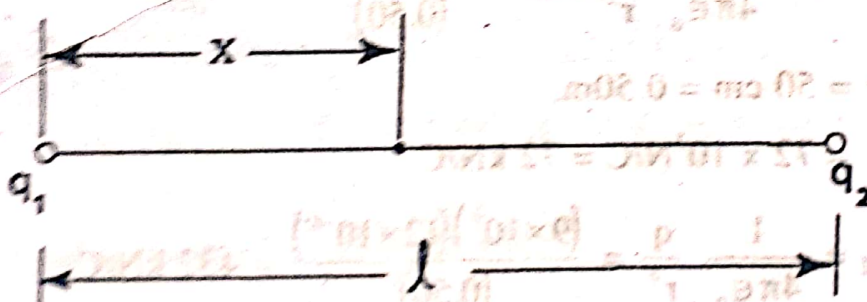
If we let $x \rightarrow 0$ in eqn. (1.15) while keeping R finite, we again obtain (1.16). This shows that at points very close to the disk, the electric field set up by the disk is the same as if the disk were infinite in extent. At points very far from the disk, the electric field produced by the disk is the same as if it were a point charge.

Example 1.12 Figure shows a charge $q_1 (= +1.0 \times 10^{-6} \text{ coul})$ 10 cm from a charge $q_2 (= +2.0 \times 10^{-6} \text{ coul})$. At what point on the line joining the two charges is the electric field strength zero?

Soln.

The point must lie between the charges because only here do the forces exerted by q_1 and q_2 on the test charge oppose each other.

Let the point P be at a distance x from q_1 . If E_1 and E_2 are the electric field strengths at P due to q_1 and q_2 respectively, we must have



$$E_1 = E_2$$

$$\text{or, } \frac{1}{4\pi\epsilon_0} \frac{q_1}{x^2} = \frac{1}{4\pi\epsilon_0} \frac{q_2}{(l-x)^2} \quad \text{where } l = 10 \text{ cm.}$$

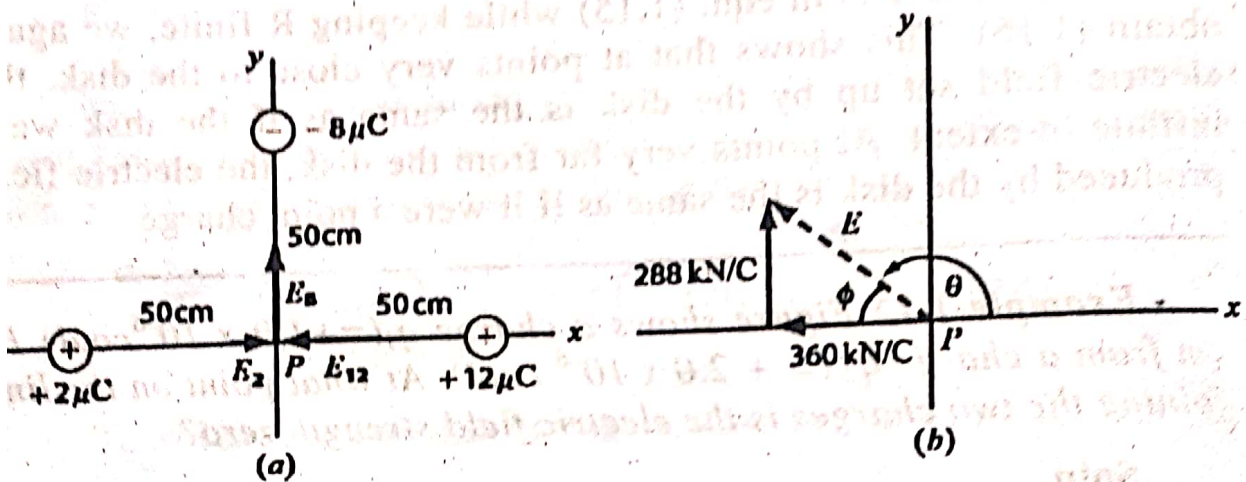
$$\text{or, } \frac{(l-x)^2}{x^2} = \frac{q_2}{q_1} = 2; \quad \text{or, } \frac{l-x}{x} = \sqrt{2}$$

$$\text{or, } \frac{10}{x} - 1 = \sqrt{2};$$

$$\text{or, } \frac{10}{x} = 1 + \sqrt{2}$$

$$\text{or, } x = \frac{10}{1 + \sqrt{2}} = 4.1 \text{ cm.}$$

Example 1.13 Find electric field at point P in figure below due to the charges shown.



Soln.

The field due to each charge are shown in the figure.

$$E_2 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{(9 \times 10^9)(2 \times 10^{-6})}{(0.50)^2}$$

where $r = 50 \text{ cm} = 0.50 \text{ m}$.

$$= 72 \times 10^3 \text{ N/C} = 72 \text{ kN/C}$$

$$E_{12} = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{(9 \times 10^9)(12 \times 10^{-6})}{(0.50)^2} = 432 \text{ kN/C}$$

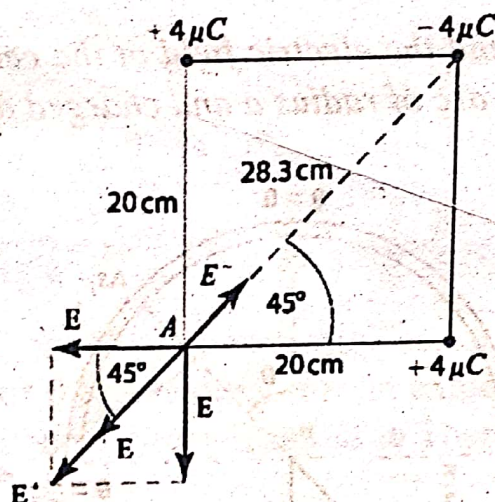
$$E_8 = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{(9 \times 10^9)(8 \times 10^{-6})}{(0.50)^2} = 288 \text{ kN/C}$$

Of these, E_2 and E_{12} act along the x -axis in opposite directions. Their resultant is $(432 - 72) = 360 \text{ kN/C}$ which act along the negative x -axis, as shown in Fig. (b). E_8 acts along the positive y -axis, also shown in the figure. Their resultant is

$$\begin{aligned} E &= (E_2 + E_{12}) + E_8 \\ &= 461 \text{ kN/C} \end{aligned}$$

The resultant field acts in a direction making an angle of 39° with the -ve x-axis [$\cos\phi = \frac{288}{360}$; $\phi = 39^\circ$].

Example 1.14 Three charges are placed at three corners of a square of side 20 cm as shown in the figure below. Find the magnitude and direction of the electric field strength at point A.



Soln.

The figure shows the fields produced at A by each of the three charges.

The fields due to $+4\mu\text{C}$ charges are

$$E = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{(9 \times 10^9)(4 \times 10^{-6})}{(0.20)^2} = 0.9 \text{ MN/C.}$$

Their directions are shown in the figure. Their resultant is

$$E^* = \sqrt{(9 \times 10^5)^2 + (9 \times 10^5)^2} = 1.27 \text{ MN/C.}$$

and its direction is as shown in the figure. The field due to the negative charge is

$$E^- = \frac{1}{4\pi\epsilon_0} \frac{q}{r^2} = \frac{(9 \times 10^9)(-4 \times 10^{-6})}{[\sqrt{(0.20)^2 + (0.20)^2}]^2} = -0.45 \text{ MN/C}$$

one, q_0 being scalar. The direction of $\mathbf{E}$ is the direction of $\mathbf{F}$, that is, it is the direction in which a stationary positive charge placed at the point would tend to move. The condition that q_0 be stationary is necessary to distinguish electric field from magnetic field as moving charges give rise to magnetic fields. It is evident from eqn. (1.5) that the unit of electric field strength is *Newton per coulomb*.

If $\mathbf{E}$ itself is produced by a set of charges *fixed in space*, then its value as given by eqn. (1.5) is independent of the size of q_0 and depends on the positions and magnitudes of other charges. On the other hand, $\mathbf{E}$ may in practice be produced by charges on bodies such as conductors. In that case, when the charge q_0 is introduced, it may cause a redistribution of the source charges and the value of $\mathbf{E}$ will then depend on the size of q_0 . While we might be interested in the field $\mathbf{E}$ for large values of q_0 , the most useful concept is that of the electric field which would exist at a point due to the original undisturbed charges. This field is defined by making the magnitude of q_0 , usually referred to as the *test charge*, as small as possible so that it does not disturb the field produced by the source charges. The electric field is, therefore, defined as

$$\mathbf{E} = \lim_{q_0 \rightarrow 0} \frac{\mathbf{F}}{q_0} \quad (1.6)$$

q_0 cannot of course be smaller than the electronic charge. Unless otherwise specified, any value quoted for $\mathbf{E}$ will be that for an undisturbed source.

Example 1.11 What is the magnitude of the electric field strength E such that an electron, placed in the field, would experience an electrical force equal to its weight?

Soln.

$$\begin{aligned} E &= \frac{F}{q_0} = \frac{mg}{q_0} \\ &= \frac{(9.1 \times 10^{-31})(9.8)}{1.6 \times 10^{-19}} \\ &= 5.6 \times 10^{-11} \text{ N/C} \end{aligned}$$

$$m = 9.1 \times 10^{-31} \text{ kg}$$

$$g = 9.8 \text{ m/s}^2$$

$$q_0 = 1.6 \times 10^{-19} \text{ C}$$

The force is repulsive.

Example 1.2 The uranium nucleus contains a charge 92 times that of a proton. If a proton is shot at the nucleus, how large a repulsive force does the proton experience due to the nucleus when it is $1 \times 10^{-11} \text{ m}$ from the nucleus centre? The nuclei of atoms are of the order of the 10^{-14} m in diameter, so the nucleus can be considered a point charge.

Soln.

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$q_1 = 92.e = (92 \times 1.6 \times 10^{-19} \text{ C})$$

$$q_2 = 1.6 \times 10^{-19} \text{ C}$$

$$= \frac{(9 \times 10^9) (92 \times 1.6 \times 10^{-19}) (1.6 \times 10^{-19})}{(1 \times 10^{-11})^2}$$

$$= 2.1 \times 10^{-4} \text{ N.}$$

• **Example 1.3** Two point charges q_1 and q_2 are 3m apart, and their combined charge is $20 \mu\text{C}$. What are the magnitude of the two charges if (a) one repels the other with a force of 0.075 N , and (b) attracts the other with a force of 0.525 N ?

Soln.

(a) Since the force is repulsive, we have from the relation

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2},$$

$$0.075 = (9 \times 10^9) \frac{q_1 q_2}{r^2}$$

$$\text{or, } q_1 q_2 = \frac{0.075 \times 9}{9 \times 10^9} = 75 \times 10^{-12} \text{ C}^2 = 75 \mu\text{C}^2.$$

$$\text{Now } q_1 + q_2 = 20 \mu\text{C},$$

$$\text{or, } q_2 = (20 - q_1)$$

$$\therefore q_1 (20 - q_1) = 75;$$

$$\text{or, } q^2 - 20q_1 + 75 = 0;$$

$$\text{or, } q_1^2 - 5q_1 - 15q_1 + 75 = 0$$

$$\text{or, } q_1 (q_1 - 5) - 15(q_1 - 5) = 0, \quad \text{or, } (q_1 - 5)(q_1 - 15) = 0$$

$$\text{or, } q_1 = 5 \quad \text{or, } 15 \mu\text{C}$$

$$\text{Hence } q_2 = 15 \quad \text{or, } 5 \mu\text{C}.$$

(b) Force is attractive; hence

$$-0.525 = (9 \times 10^9) \frac{q_1 q_2}{9}$$

$$\text{or, } q_1 q_2 = \frac{(-0.525)(9)}{9 \times 10^9} = -525 \mu\text{C}^2.$$

Again substituting $q_2 = (20 - q_1)$

$$q_1(20 - q_1) = 525$$

$$\text{or, } q_1^2 - 20q_1 - 525 = 0$$

$$\text{or, } q_1^2 + 15q_1 - 35q_1 - 525 = 0$$

$$\text{or, } q_1(q_1 + 15) - 35(q_1 + 15) = 0$$

$$\text{or, } (q_1 + 15)(q_1 - 35) = 0$$

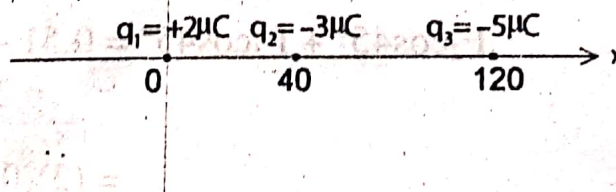
$$\text{or, } q_1 = -15 \quad \text{or, } +35 \mu\text{C} \quad \text{and} \quad q_2 = +35 \quad \text{or, } -15 \mu\text{C}.$$

Example 1.4 Three point charges are placed at the following points on the x-axis: $+2 \mu\text{C}$ at $x = 0$, $-3 \mu\text{C}$ at $x = 40\text{cm}$, $-5 \mu\text{C}$ at $x = 120\text{cm}$. Find the force on the $-3 \mu\text{C}$ charge.

Soln.

The arrangement is shown in the adjoining figure. The force on q_2 is the vector sum of two contributions, the attractive force due to q_1 and the repulsive force due to q_3 .

The direction of both the forces is towards the left i.e., towards q_1 . The sum of these two forces, taken algebraically, since they are along the same line, is

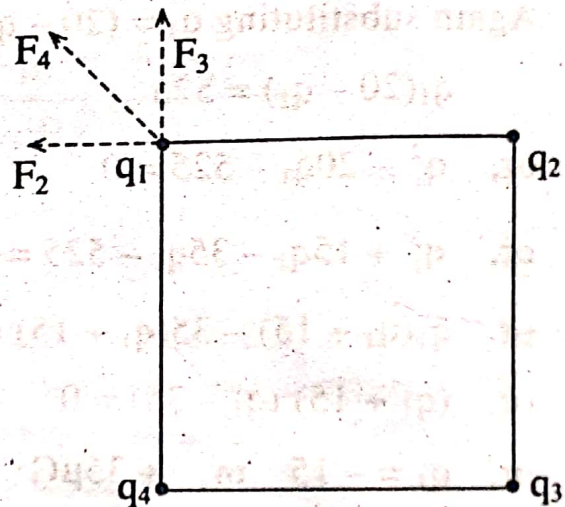


$$\begin{aligned}
 F &= F_1 + F_2 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_1^2} + \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r_2^2} \\
 &= 9 \times 10^9 \left[\frac{(2 \times 10^{-6})(3 \times 10^{-6})}{(0.40)^2} + \frac{(5 \times 10^{-6})(3 \times 10^{-6})}{(0.80)^2} \right] \\
 &= 0.55 \text{ N towards left.}
 \end{aligned}$$

Example 1.5 Four equal point charges, $+3\text{mC}$, are placed at the four corners of a square that is 40 cm on a side. Find the force on any one of the charges.

Soln.

The situation is depicted in the adjoining figure. Let us consider the forces acting on q_1 . Let F_2 , F_3 and F_4 be the forces exerted on q_1 by q_2 , q_3 and q_4 respectively.



By symmetry

$$F_2 = F_3 = (9 \times 10^9) \frac{(3 \times 10^{-6})(3 \times 10^{-6})}{(0.40)^2} = 0.51 \text{ N.}$$

Since the directions of F_2 and F_3 are along the edges as shown in the figure, their vector sum will lie along the diagonal from q_4 to q_1 and have the magnitude.

$$\begin{aligned}
 F_2 \cos 45^\circ + F_3 \cos 45^\circ &= 0.51 \frac{1}{\sqrt{2}} + 0.51 \frac{1}{\sqrt{2}} \\
 &= (2)(0.51) \frac{1}{\sqrt{2}} = 0.72 \text{ N.}
 \end{aligned}$$

The remaining force F_4 is also along this diagonal, and

$$F_4 = (9 \times 10^9) \frac{(3 \times 10^{-6})(3 \times 10^{-6})}{(\sqrt{0.40^2 + 0.40^2})^2} = 0.25 \text{ N}$$

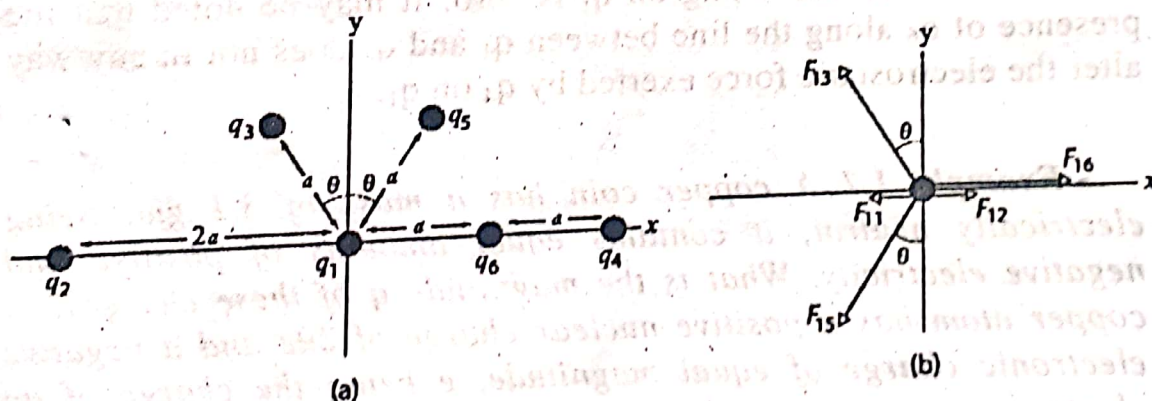
The resultant of all the three force acting along the diagonal away from the square.

$$= 0.72 \text{ N} + 0.25 \text{ N} = 0.97 \text{ N}.$$

Example 1.6 The figure below shows an arrangement of six fixed charged particles, where $a = 2.0 \text{ cm}$ and $\theta = 30^\circ$. All six particles have the same magnitude of charge $q = 3.0 \times 10^{-6} \text{ C}$; their electrical signs are as indicated. What is the net electrostatic force F_1 acting on q_1 due to other charges?

Soln.

From the figure it is obvious that F_1 , the net electrostatic force acting on q_1 is the vector sum of F_{12} , F_{13} , F_{14} , F_{15} and F_{16} , the exerted on q_1 due to other charges. Because q_2 and q_4 are equal in magnitude and are both a distance $r = 2a$ from q_1 , we have from eqn. (1.4)



$$F_{12} = F_{14} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{(2a)^2} \quad (i)$$

Similarly, since q_3 , q_5 and q_6 are equal in magnitude and are each a distance $r = a$ from q_1 , we have

$$F_{13} = F_{15} = F_{16} = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{a^2} \quad (ii)$$

Fig. (b) is a free body diagram for q_1 . It and eqn. (i) show that F_{12} and F_{14} are equal in magnitude but opposite in direction, thus they cancel each other. Inspection of the figure (b) and eqn. (ii) reveals that the y-components of F_{13} and F_{15} also cancel and that their x-components are identical in magnitude and both point in the direction of decreasing x. Fig. (b) also shows that F_{16} points in the direction of increasing x. Thus F_1 must be parallel to the x-axis; its magnitude is the difference between F_{16} and twice the x-component of F_{13} .

$$\begin{aligned}\therefore F_1 &= F_{16} - 2F_{13} \sin \theta \\ &= \frac{1}{4\pi\epsilon_0} \frac{q_1 q_6}{a^2} - 2 \cdot \frac{1}{4\pi\epsilon_0} \frac{q_1 q_3}{a^2} \sin \theta\end{aligned}$$

Now $q_3 = q_6$ and $\theta = 30^\circ$,

Substituting these values we find

$$F_1 = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_6}{a^2} - 2 \cdot \frac{1}{4\pi\epsilon_0} \frac{q_1 q_6}{a^2} \sin 30^\circ = 0$$

So the net force acting on q_1 is zero. It may be noted that the presence of q_6 along the line between q_1 and q_4 does not in any way alter the electrostatic force exerted by q_4 on q_1 .

Example 1.7 A copper coin has a mass of 3.1 gm. Being electrically neutral, it contains equal amounts of positive and negative electricity. What is the magnitude q of these charges? A copper atom has a positive nuclear charge of $29e$ and a negative electronic charge of equal magnitude, e being the charge of an electron.

Soln.

The number N of copper atoms in the coin

$$\begin{aligned}N &= \frac{m}{M} \cdot N_0 \quad \text{where } m = \text{mass of the coin} \\ & \quad M = \text{atomic weight of copper} \\ &= \frac{3.1}{64} \times 6.0 \times 10^{23} \quad N_0 = \text{Avogadro's number}\end{aligned}$$

$$= 2.9 \times 10^{22} \text{ atoms.}$$

Charge (either positive or negative) of a copper atom

$$= 29 \times 1.6 \times 10^{-19} \text{ C} = 4.64 \times 10^{-18} \text{ C.}$$

Therefore, the charge

$$q = (4.64 \times 10^{-18}) (2.9 \times 10^{22}) = 1.345 \times 10^5 \text{ C.}$$

This is an enormous amount of charge. It can be verified that it will take 40 hrs. for a charge of this amount to pass through a 100 watt, 110 volt light bulb.

Example 1.8 What should be the distance of separation between the total positive and negative charges of the copper coin of example 1.7 such that their force of attraction is 4.5N?

Soln.

$$F = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$q_1 = q_2 = 1.345 \times 10^5 \text{ C.}$$

$$F = 4.5 \text{ N}$$

$$= \frac{1}{4\pi\epsilon_0} \frac{q^2}{r^2}$$

$$\therefore r = q \cdot \sqrt{\frac{1/4\pi\epsilon_0}{F}} = (1.345 \times 10^5) \sqrt{\frac{9 \times 10^9}{4.5}}$$

$$= (1.345 \times 10^5) (0.4472 \times 10^5)$$

$$= 6.0 \times 10^9 \text{ metres}$$

$$= 3.75 \times 10^6 \text{ miles.}$$

Example 1.9 The distance r between the electron and the proton in the hydrogen atom is about 5.3×10^{-11} metre. What are the magnitudes of (a) the electrical force and (b) the gravitational force between these two particles?

Soln.

$$F_e = \frac{1}{4\pi\epsilon_0} \frac{q_1 q_2}{r^2}$$

$$q_1 = q_2 = 1.6 \times 10^{-19} \text{ C}$$

$$r = 5.3 \times 10^{-11} \text{ m.}$$

$$= (9 \times 10^9) \frac{(1.6 \times 10^{-19})^2}{(5.3 \times 10^{-11})^2}$$

$$= 8.1 \times 10^{-8} \text{ N.}$$

The gravitational force is given by

$$F_g = G \frac{m_1 m_2}{r^2}$$

$$= \frac{(6.7 \times 10^{-11})(9.1 \times 10^{-31})(1.7 \times 10^{-27})}{(5.3 \times 10^{-11})^2}$$

$$= 3.7 \times 10^{-47} \text{ N.}$$

m_1 = electron mass
 $= 9.1 \times 10^{-31} \text{ Kg}$

m_2 = proton mass
 $= 1.7 \times 10^{-27} \text{ Kg}$

G = gravitational constant
 $= 6.7 \times 10^{-11} \text{ N-m}^2/\text{kg}^2$

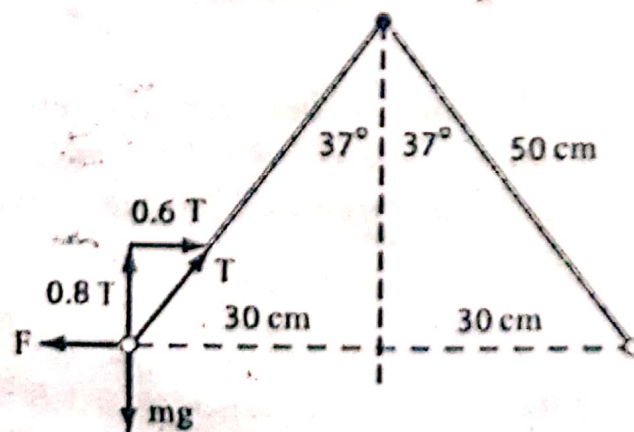
r = $5.3 \times 10^{-11} \text{ m.}$

Thus the electrical force is approximately 2.2×10^{39} times stronger than the gravitational force.

Example 1.10 The two balls shown in the figure below have identical masses of 0.20gm each. When suspended from 50cm long strings, they make an angle of 37° to the vertical. If the charges on each are the same, how large is each charge?

Soln.

Since the system is at rest, let us apply the conditions for equilibrium to the ball on the left. Three forces act on this ball: its weight mg ; the tension T in the string; and F , the repulsive force due to the charge on the other ball.



Applying the usual conditions for equilibrium:

$$\sum F_x = 0 ; \quad \text{i.e., } F - 0.6T = 0 \quad (i)$$

$$\sum F_y = 0 ; \quad \text{i.e., } 0.8T - mg = 0$$

$$\text{or, } 0.8T - (0.2 \times 10^{-3} \text{kg}) (9.8 \text{m/s}^2) = 0$$

$$\text{or, } T = 2.45 \times 10^{-3} \text{N} \quad (ii)$$

Substituting the value of T as given by (ii) in (i) we get

$$F - (0.6) (2.45 \times 10^{-3} \text{N}) = 0$$

$$\text{or, } F = 1.47 \times 10^{-3} \text{N} \quad (iii)$$

Substituting this value of F in Coulomb's law

$$1.47 \times 10^{-3} = (9 \times 10^9) \frac{q^2}{(0.60)^2}$$

$$\text{or, } q^2 = \frac{(1.47 \times 10^{-3})(0.60)^2}{9 \times 10^9} = 0.0588 \times 10^{-12}$$

$$= 5.88 \times 10^{-14}$$

$$\text{or, } q = 2.4248 \times 10^{-7} \text{C}$$

$$= 0.24248 \times 10^{-6} \text{C}$$

$$= 0.24248 \mu\text{C}.$$

1.3 Electric field and electric field strength

Suppose that a stationary positive charge q_0 , placed at any point in a certain region of space, experiences a force F . We then say there is an *electric field* in the region, whose strength E at the point is given by

$$E = \frac{F}{q_0} \quad (1.5)$$

The *electric field strength* E , also known as *electric intensity*, at a point is; therefore, defined as the force per unit positive charge (assumed positive for convenience). Here E is a vector because F is